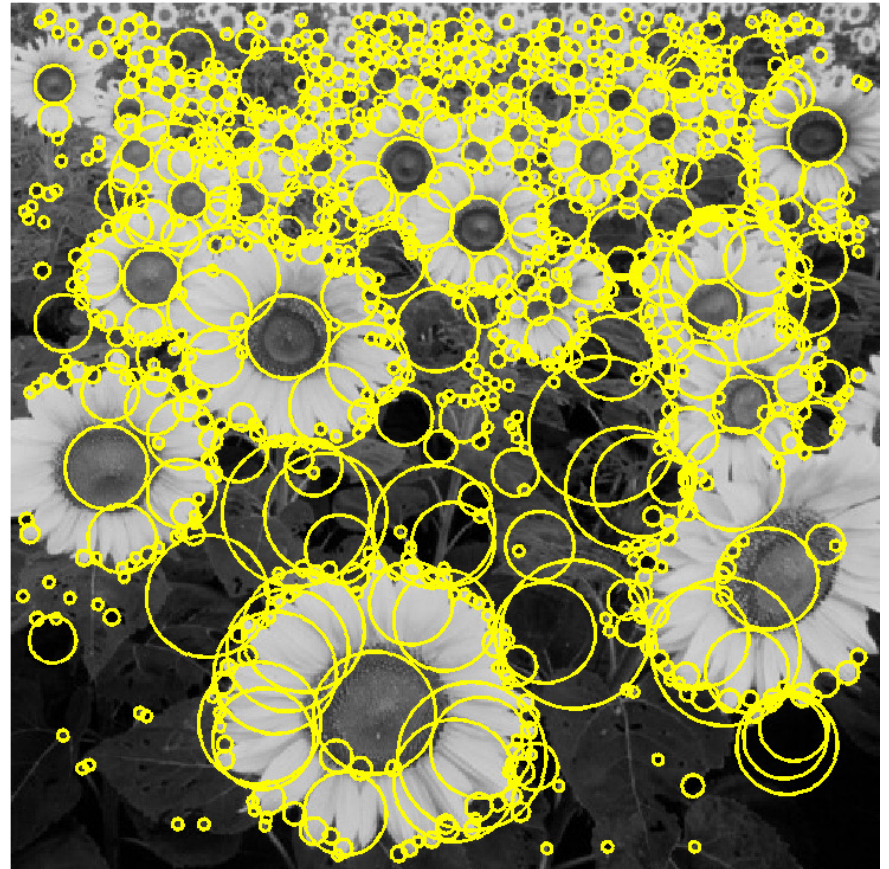


Harris corner detection

Feature extraction: Corners and blobs

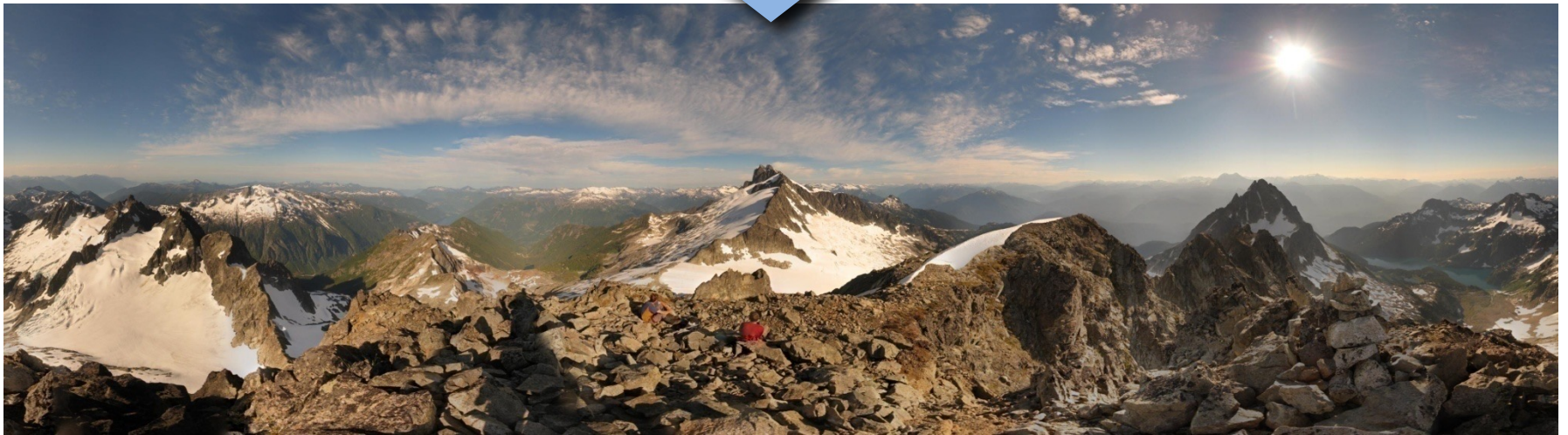
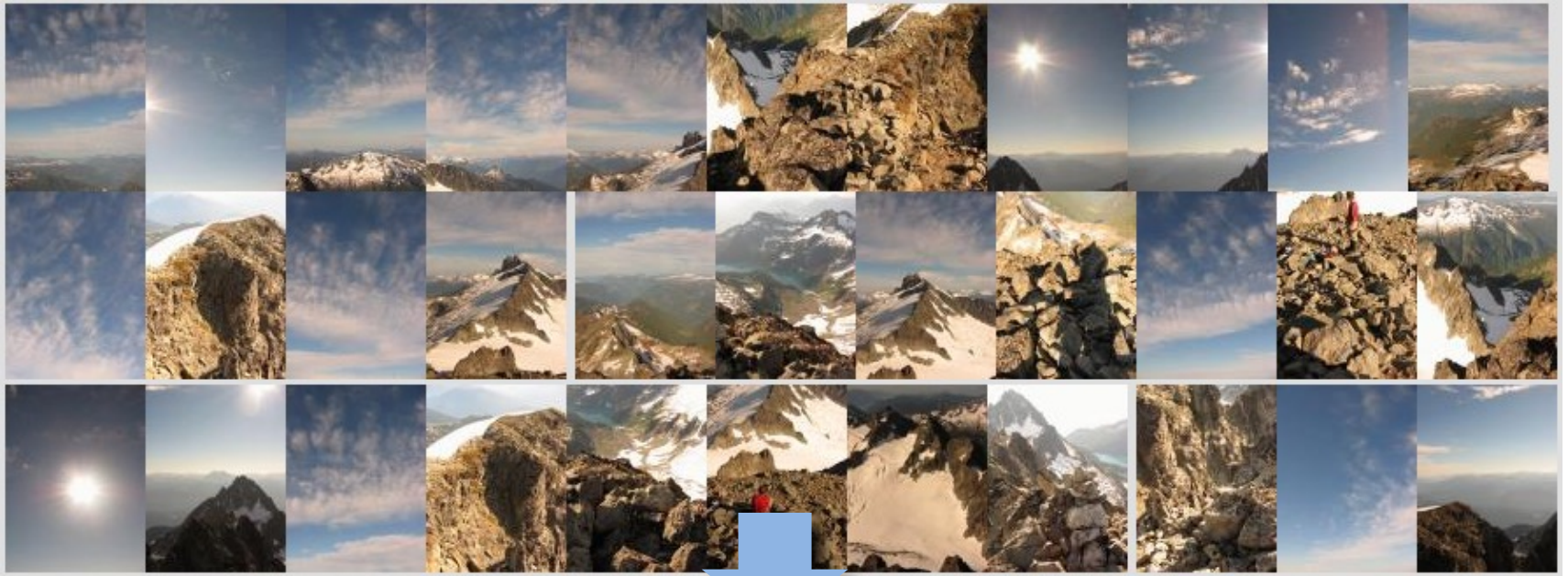


Feature extraction: Corners and blobs

- Interest points:
 - A corner can be defined as the intersection of two edges.
 - blob detection methods are aiming at detecting regions in the image that differ in properties, such as brightness or color, compared to surrounding regions

(credit: Wikipedia)

Motivation: Automatic panoramas



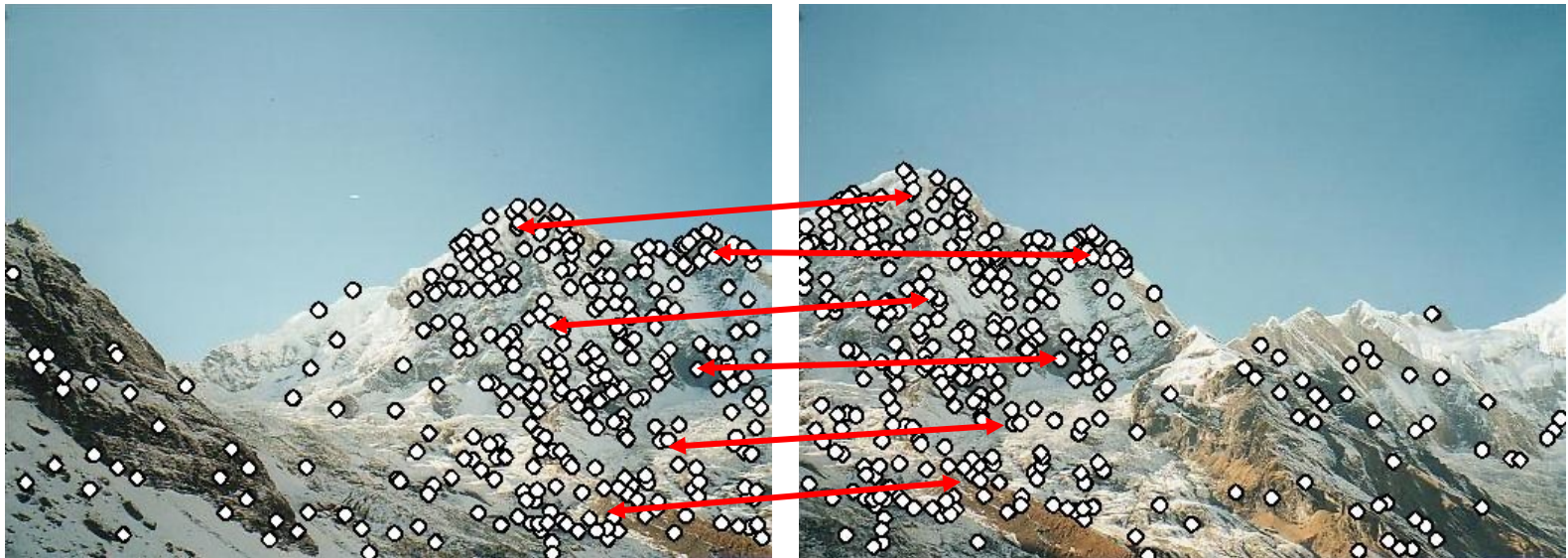
Why extract features?

- Motivation: panorama stitching
 - We have two images – how do we combine them?



Why extract features?

- Motivation: panorama stitching
 - We have two images – how do we combine them?



Step 0: interest point detection

Step 1: extract features

Step 2: match features

Why extract features?

- Motivation: panorama stitching
 - We have two images – how do we combine them?



Step 0: interest point detection

Step 1: extract features

Step 2: match features

Step 3: align images

Image matching



by [Diva Sian](#)



by [swashford](#)

Harder case

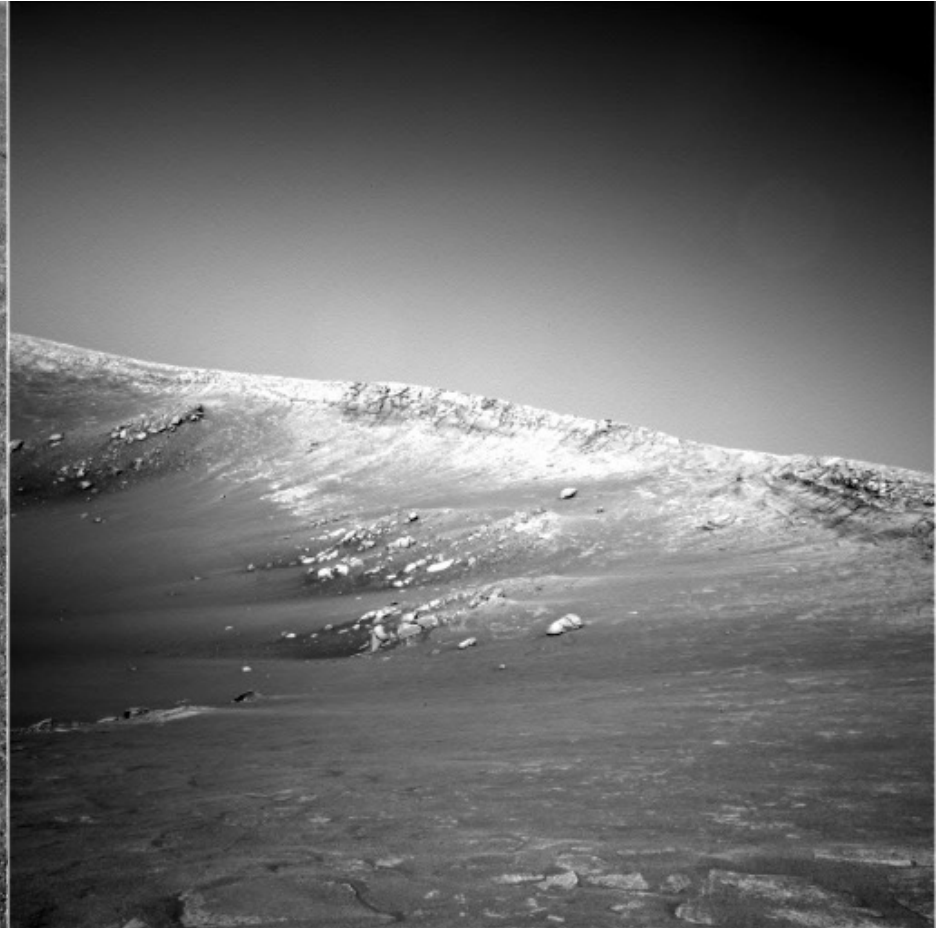


by [Diva Sian](#)

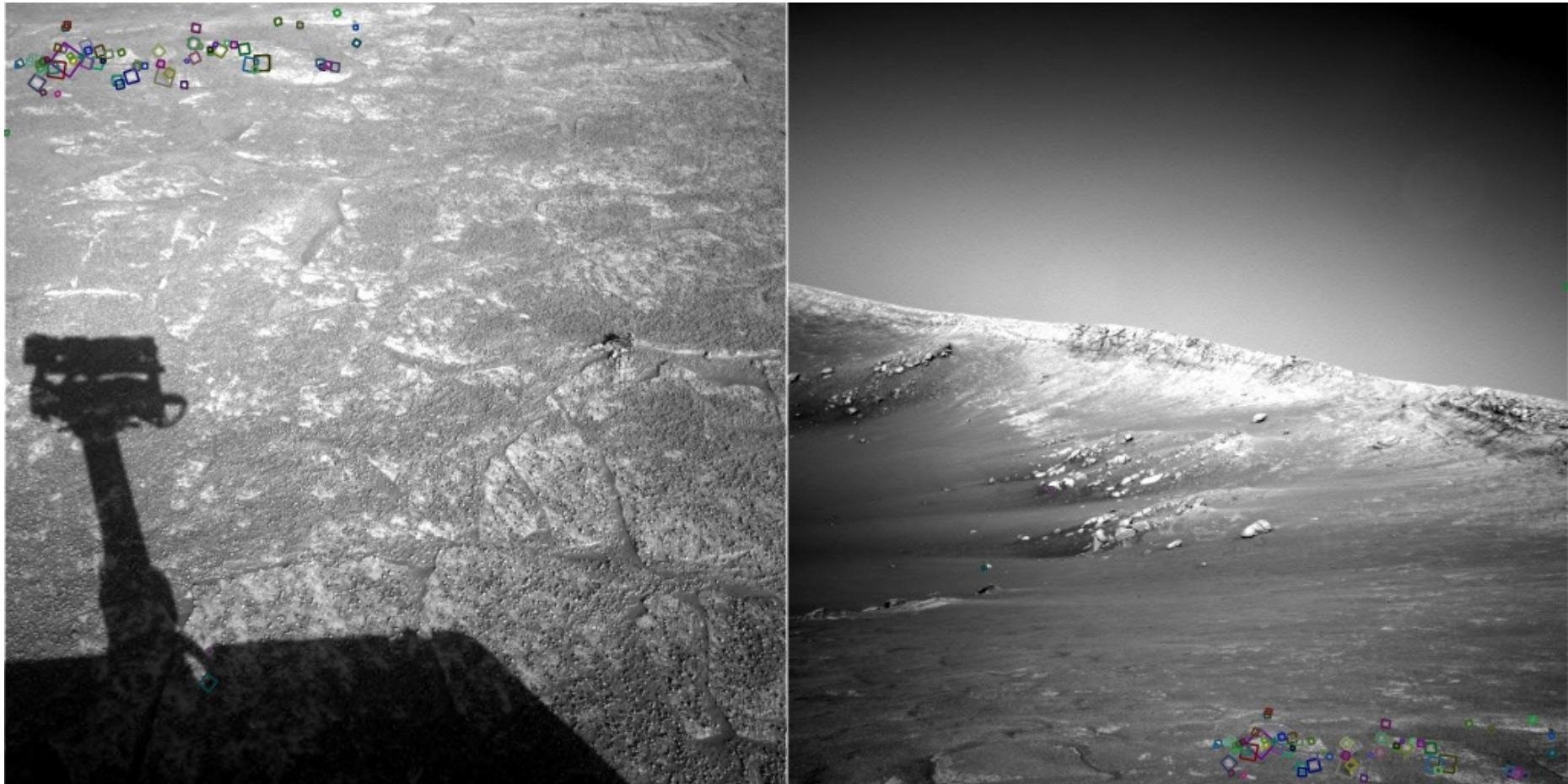


by [scgbt](#)

Harder still?

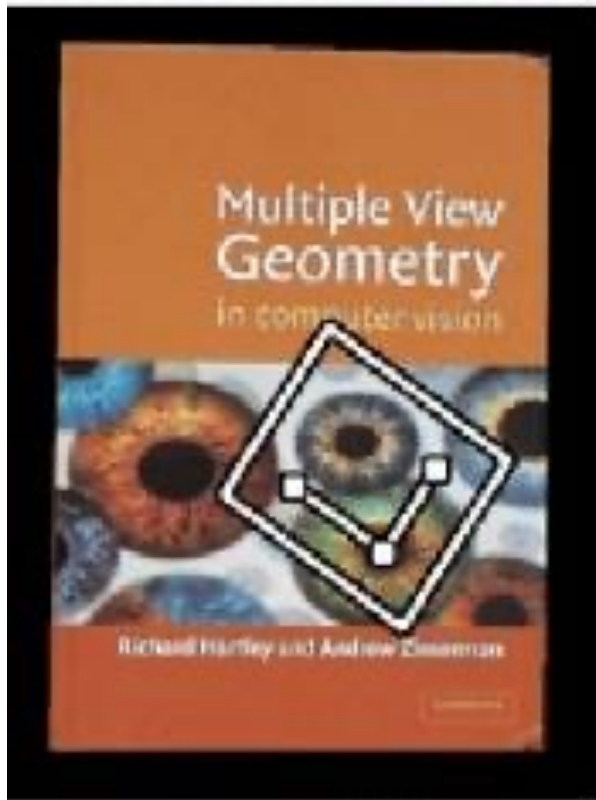


Answer below (look for tiny colored squares...)



NASA Mars Rover images
with SIFT feature matches

Feature Matching



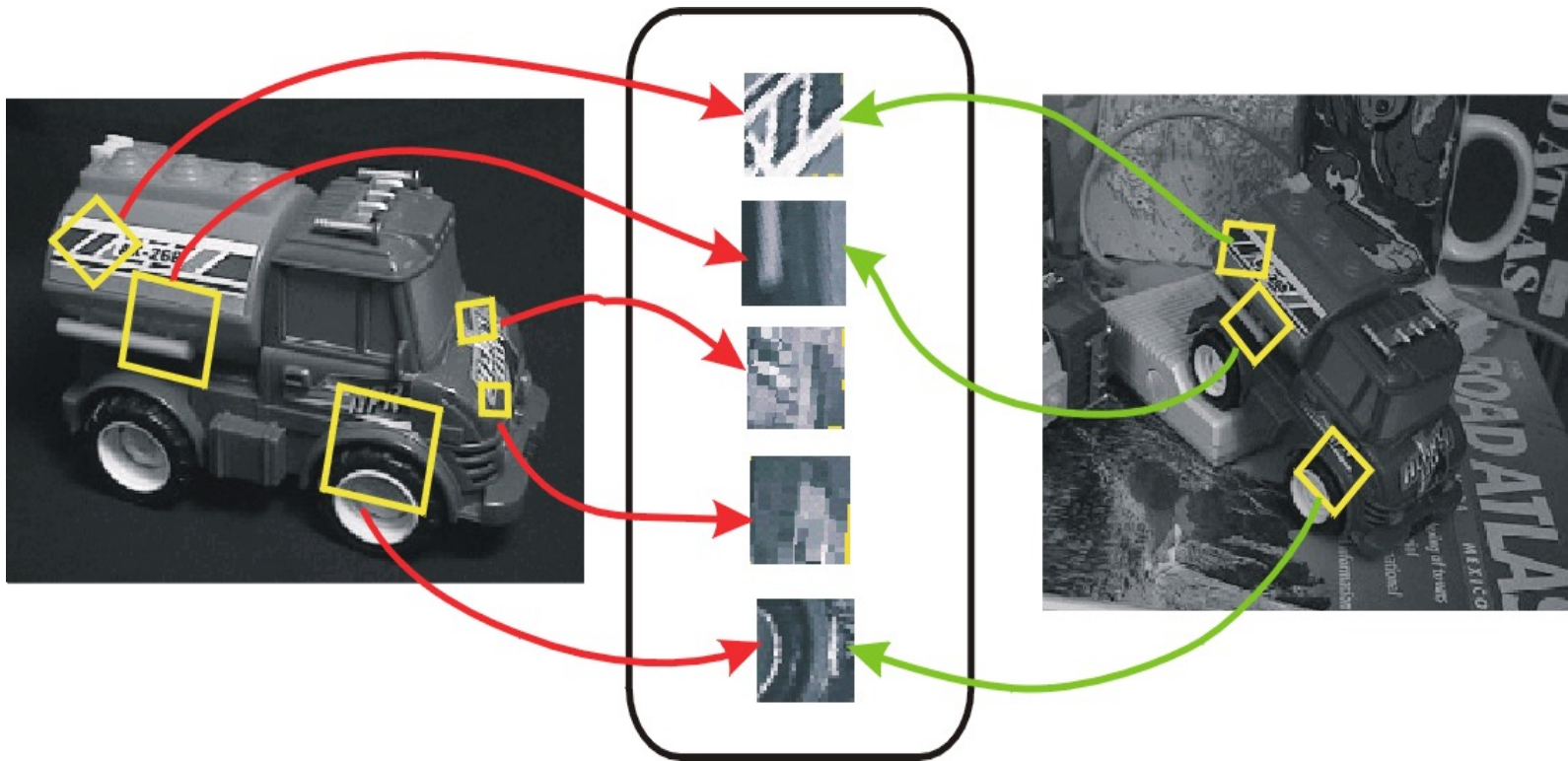
Feature Matching



Invariant local features

Find features that are invariant to transformations

- geometric invariance: translation, rotation, scale
- photometric invariance: brightness, exposure, ...



Feature Descriptors

Advantages of local features

Locality

- features are local, so robust to occlusion and clutter

Quantity

- hundreds or thousands in a single image

Distinctiveness:

- can differentiate a large database of objects

Efficiency

- real-time performance achievable

More motivation...

Feature points are used for:

- Image alignment
- 3D reconstruction
- Motion tracking
- Object recognition
- Indexing and database retrieval
- Robot navigation
- ... other

Approach

Feature detection: find it

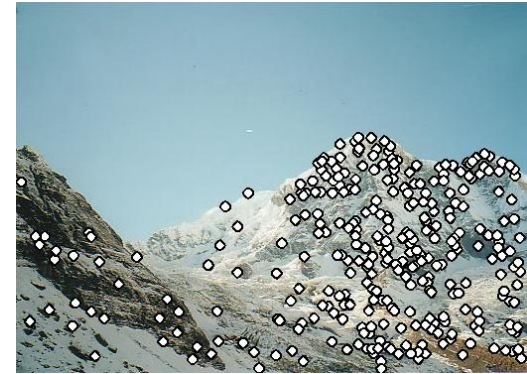
Feature descriptor: represent it

Feature matching: match it

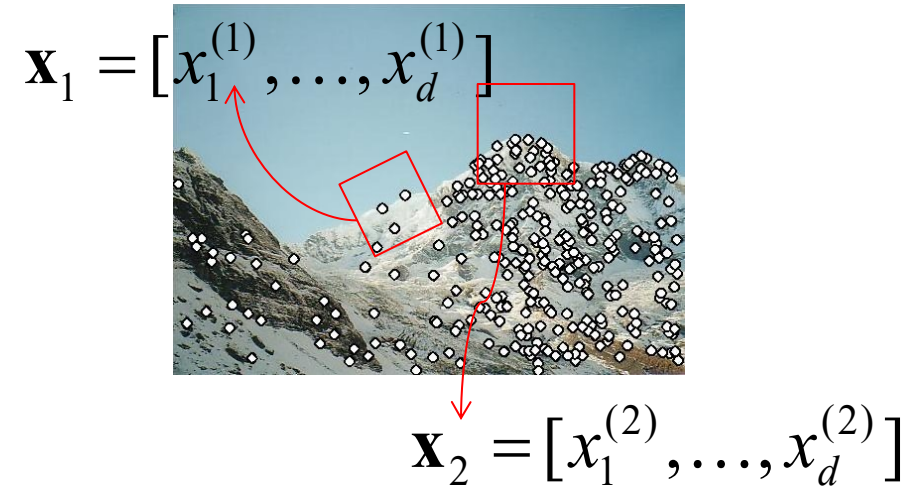
Feature tracking: track it, when motion

Local features: main components

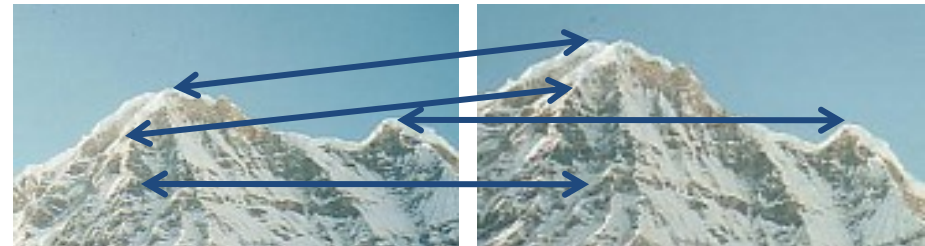
1) Detection: Identify the interest points



2) Description: Extract vector feature descriptor surrounding each interest point.



3) Matching: Determine correspondence between descriptors in two views



What makes a good feature?



Want uniqueness

Look for image regions that are unusual

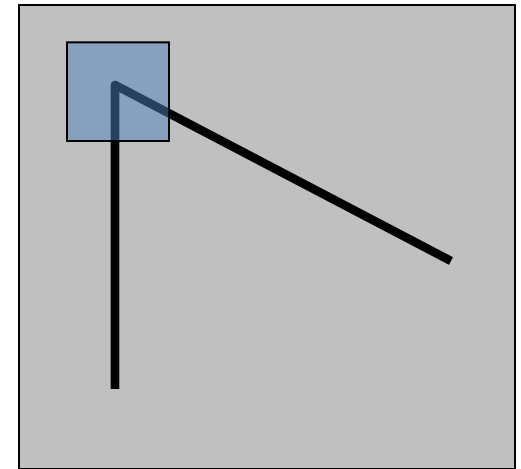
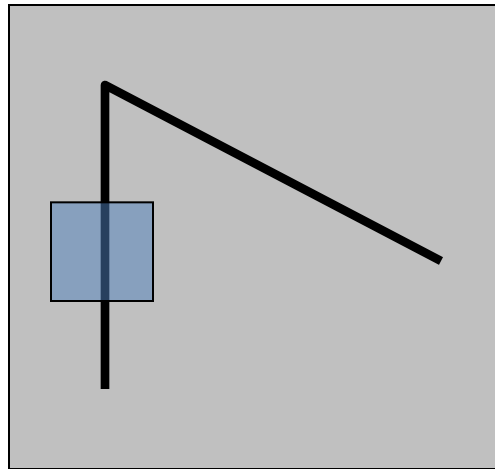
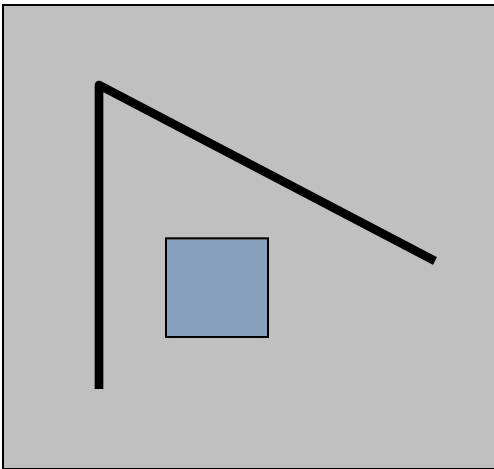
- Lead to distinct matches in other images
- The particular object in only a few number of images

How to define “unusual”?

Local measures of uniqueness

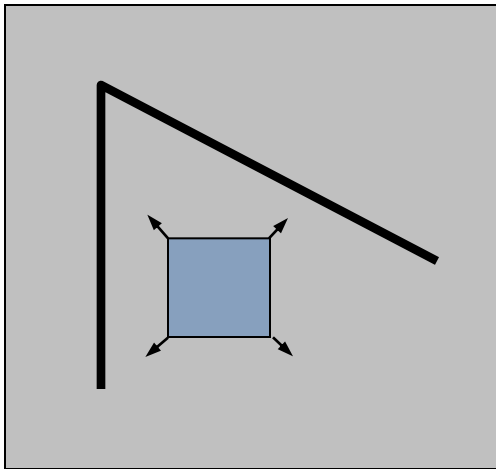
Suppose we only consider a small window of pixels

- What defines whether a feature is a good or bad candidate?

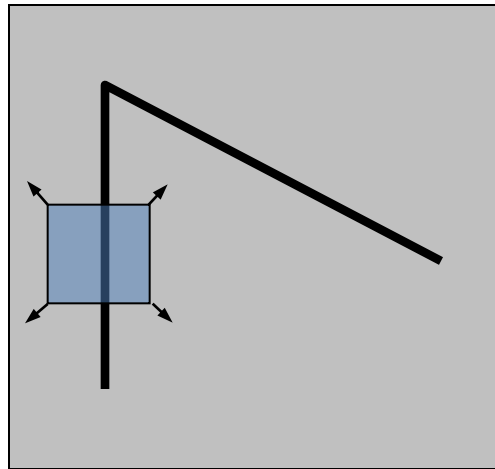


Local measure of feature uniqueness

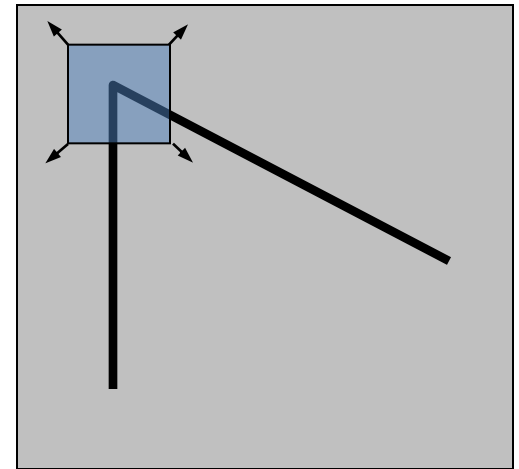
- How does the window change when you shift it?
- Shifting the window in any direction causes a big change



“flat” region:
no change in all
directions



“edge”:
no change along the
edge direction



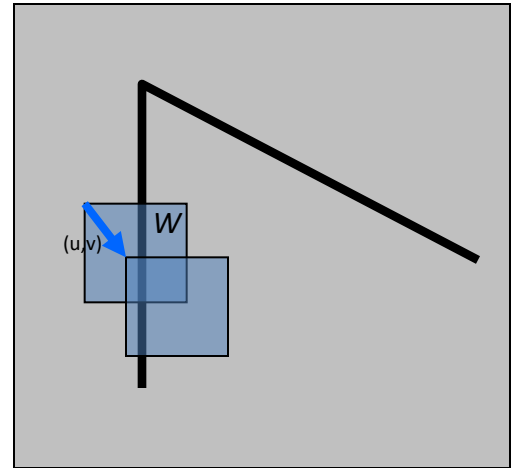
“corner”:
significant change in
all directions

Harris corner detection: the math

Consider shifting the window W by (u, v)

- how do the pixels in W change?
- compare each pixel before and after by summing up the squared differences (SSD)
- this defines an SSD “error” $E(u, v)$:

$$E(u, v) = \sum_{(x, y) \in W} [I(x + u, y + v) - I(x, y)]^2$$



- We are happy if this error is high
- Slow to compute exactly for each pixel and each offset (u, v)

Small motion assumption

Taylor Series expansion of I :

$$I(x+u, y+v) = I(x, y) + \frac{\partial I}{\partial x}u + \frac{\partial I}{\partial y}v + \text{higher order terms}$$

If the motion (u,v) is small, then first order approximation is good

$$\begin{aligned} I(x+u, y+v) &\approx I(x, y) + \frac{\partial I}{\partial x}u + \frac{\partial I}{\partial y}v \\ &\approx I(x, y) + [I_x \ I_y] \begin{bmatrix} u \\ v \end{bmatrix} \end{aligned}$$

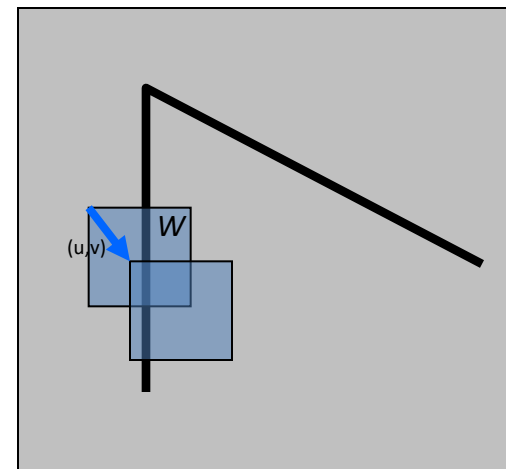
shorthand: $I_x = \frac{\partial I}{\partial x}$

Plugging this into the formula on the previous slide...

Corner detection: the math

Consider shifting the window W by (u, v)

- define an SSD “error” $E(u, v)$:



$$\begin{aligned} E(u, v) &= \sum_{(x, y) \in W} [I(x + u, y + v) - I(x, y)]^2 \\ &\approx \sum_{(x, y) \in W} [I(x, y) + I_x u + I_y v - I(x, y)]^2 \\ &\approx \sum_{(x, y) \in W} [I_x u + I_y v]^2 \end{aligned}$$

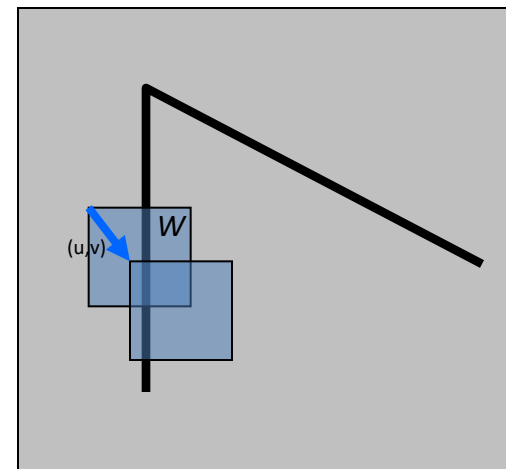
Corner detection: the math

Consider shifting the window W by (u, v)

- define an SSD “error” $E(u, v)$:

$$\begin{aligned} E(u, v) &\approx \sum_{(x, y) \in W} [I_x u + I_y v]^2 \\ &\approx Au^2 + 2Buv + Cv^2 \end{aligned}$$

$$A = \sum_{(x, y) \in W} I_x^2 \quad B = \sum_{(x, y) \in W} I_x I_y \quad C = \sum_{(x, y) \in W} I_y^2$$



Corner detection: the math

The surface $E(u,v)$ is locally approximated:

$$E(u, v) \approx Au^2 + 2Buv + Cv^2$$
$$\approx \begin{bmatrix} u & v \end{bmatrix} \underbrace{\begin{bmatrix} A & B \\ B & C \end{bmatrix}}_H \begin{bmatrix} u \\ v \end{bmatrix}$$
$$A = \sum_{(x,y) \in W} I_x^2$$

$$B = \sum_{(x,y) \in W} I_x I_y$$

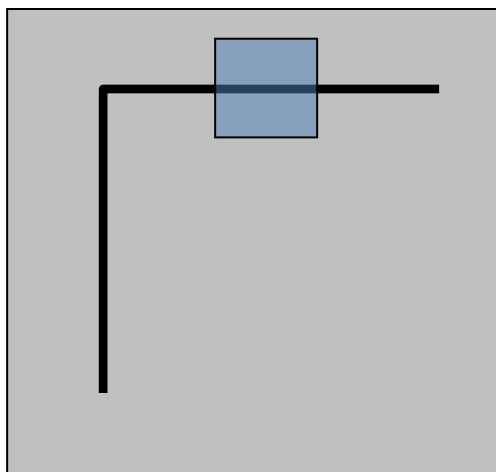
$$C = \sum_{(x,y) \in W} I_y^2$$

$$E(u, v) \approx \begin{bmatrix} u & v \end{bmatrix} \underbrace{\begin{bmatrix} A & B \\ B & C \end{bmatrix}}_H \begin{bmatrix} u \\ v \end{bmatrix}$$

$$A = \sum_{(x,y) \in W} I_x^2$$

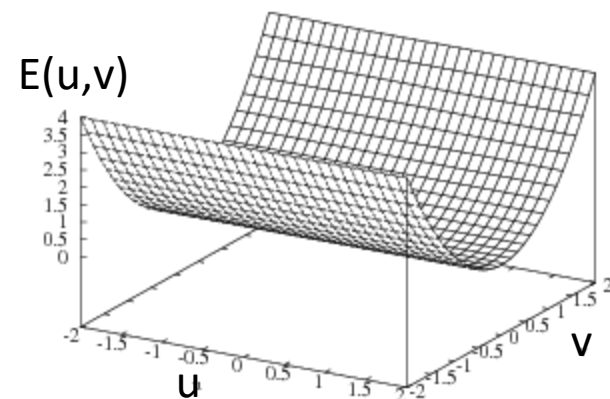
$$B = \sum_{(x,y) \in W} I_x I_y$$

$$C = \sum_{(x,y) \in W} I_y^2$$



Horizontal edge: $I_x = 0$

$$H = \begin{bmatrix} 0 & 0 \\ 0 & C \end{bmatrix}$$

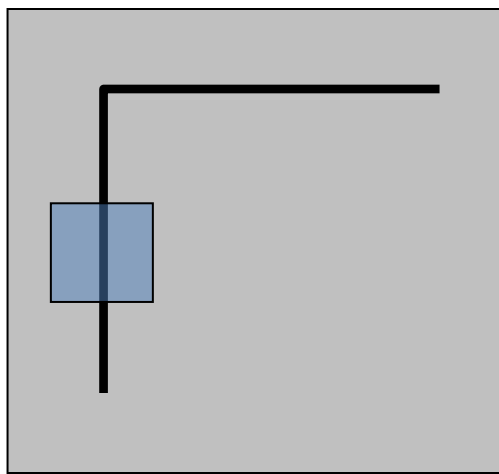


$$E(u, v) \approx \begin{bmatrix} u & v \end{bmatrix} \underbrace{\begin{bmatrix} A & B \\ B & C \end{bmatrix}}_H \begin{bmatrix} u \\ v \end{bmatrix}$$

$$A = \sum_{(x,y) \in W} I_x^2$$

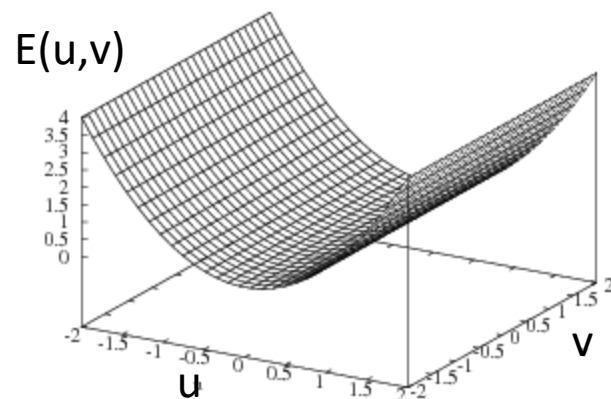
$$B = \sum_{(x,y) \in W} I_x I_y$$

$$C = \sum_{(x,y) \in W} I_y^2$$



Vertical edge: $I_y = 0$

$$H = \begin{bmatrix} A & 0 \\ 0 & 0 \end{bmatrix}$$



Quick eigenvalue/eigenvector review

The **eigenvectors** of a matrix **A** are the vectors **x** that satisfy:

$$Ax = \lambda x$$

The scalar λ is the **eigenvalue** corresponding to **x**

- The eigenvalues are found by solving:

$$\det(A - \lambda I) = 0$$

- In our case, **A** = **H** is a 2x2 matrix, so we have

$$\det \begin{bmatrix} h_{11} - \lambda & h_{12} \\ h_{21} & h_{22} - \lambda \end{bmatrix} = 0$$

- The solution:

$$\lambda_{\pm} = \frac{1}{2} \left[(h_{11} + h_{22}) \pm \sqrt{4h_{12}h_{21} + (h_{11} - h_{22})^2} \right]$$

Example

$$P^T = A = \begin{bmatrix} 0.1 & 0.3 \\ 0.9 & 0.7 \end{bmatrix}$$

Solve this equation:

X is eigenvector

λ contains eigenvalues

$$\begin{aligned} AX &= \lambda X \\ (A - \lambda I)X &= 0 \end{aligned}$$

By:

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 0.1 - \lambda & 0.3 \\ 0.9 & 0.7 - \lambda \end{vmatrix} = 0$$

$$(0.1 - \lambda)(0.7 - \lambda) - 0.9 \times 0.3 = 0$$

$$\lambda^2 - 0.8\lambda - 0.2 = 0$$

$$\lambda = 1 \text{ or } \lambda = -0.2$$

Quiz

- Which of the following can be called as the “corners”
 - Pixels with large gradient magnitude
 - Pixels at the end of a straight line
 - Pixels at the junction of horizontal and vertical lines

Interpreting the eigenvalues

Classification of image points using eigenvalues of H :

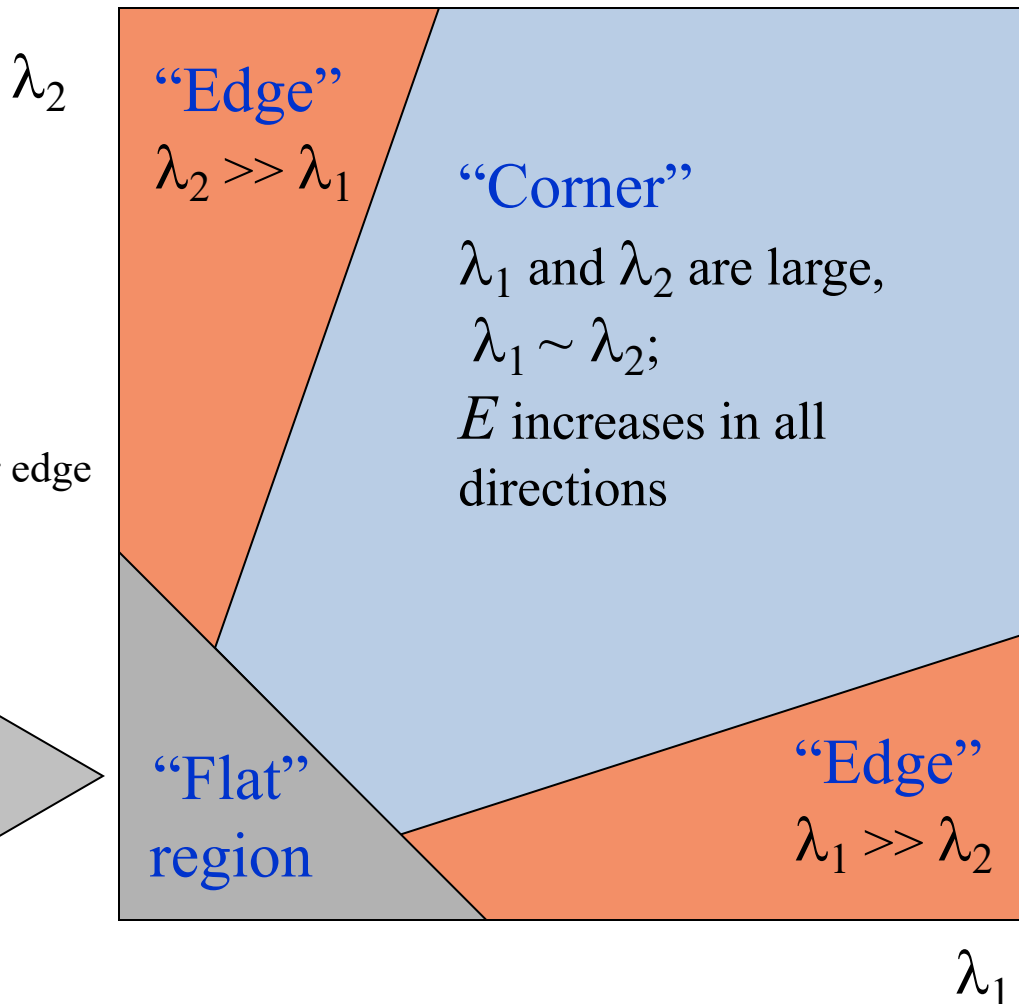
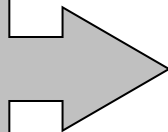
$$R = \lambda_1 \lambda_2 - k(\lambda_1 + \lambda_2)^2$$

R is large for corner

R is negative (with large magnitude) for edge

R is small for flat region

λ_1 and λ_2 are small;
 E is almost constant
in all directions



Other Versions of The Harris operator

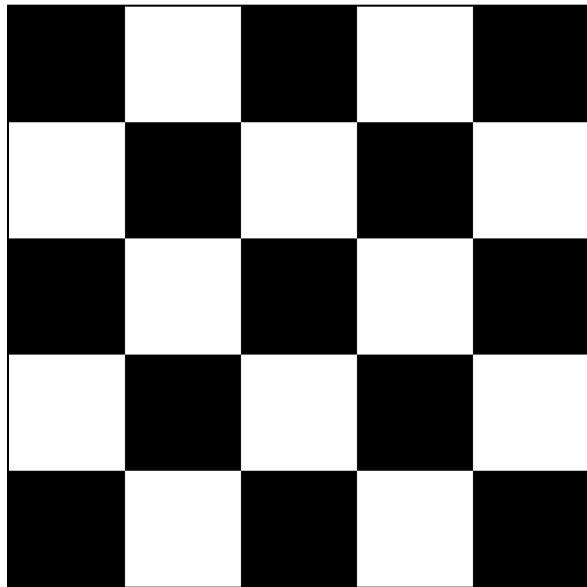
$$\begin{aligned} f &= \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2} \\ &= \frac{\textit{determinant}(H)}{\textit{trace}(H)} \end{aligned}$$

$\lambda_{\min}$ is a variant of the “Harris operator” for feature detection

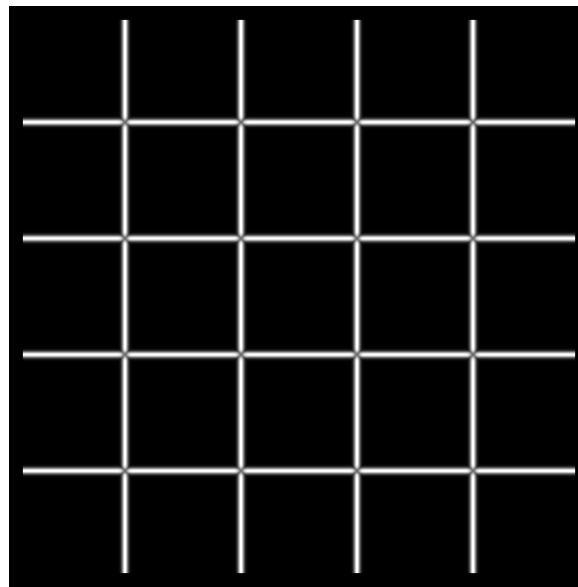
Corner detection summary

Here's what you do

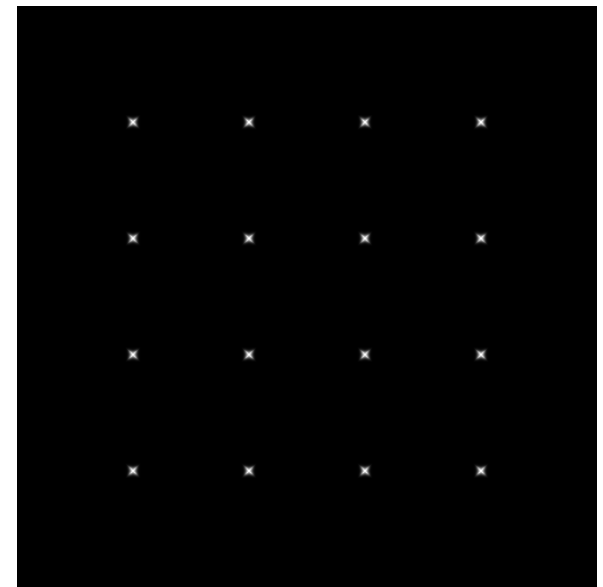
- Compute the gradient at each point in the image
- Create the H matrix from the entries in the gradient
- Compute the eigenvalues.
- Find points with large response ($\lambda_{\min} > \text{threshold}$)
- Choose those points where $\lambda_{\min}$ is a local maximum as features



I



$\lambda_{\max}$

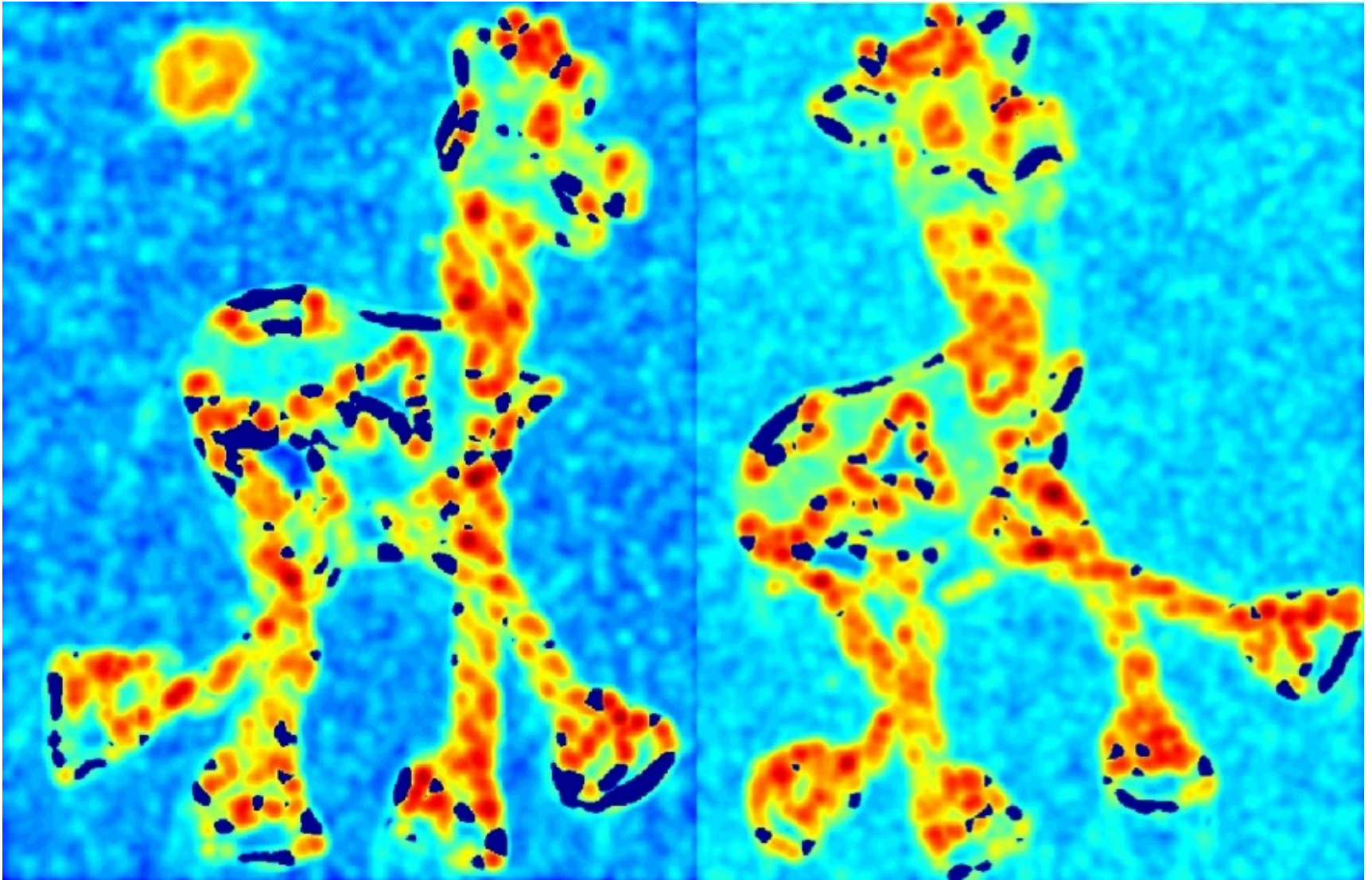


$\lambda_{\min}$

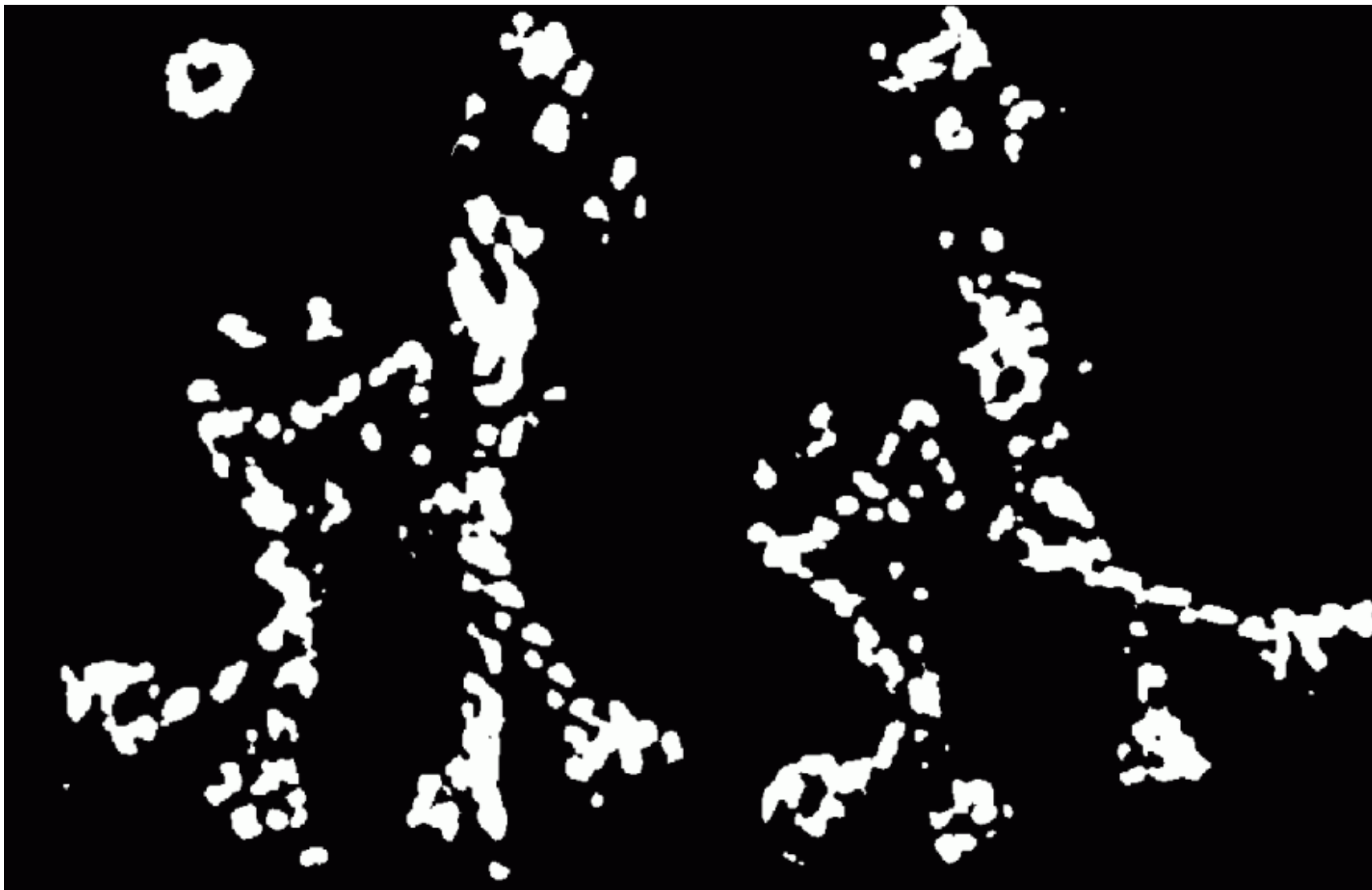
Harris detector example



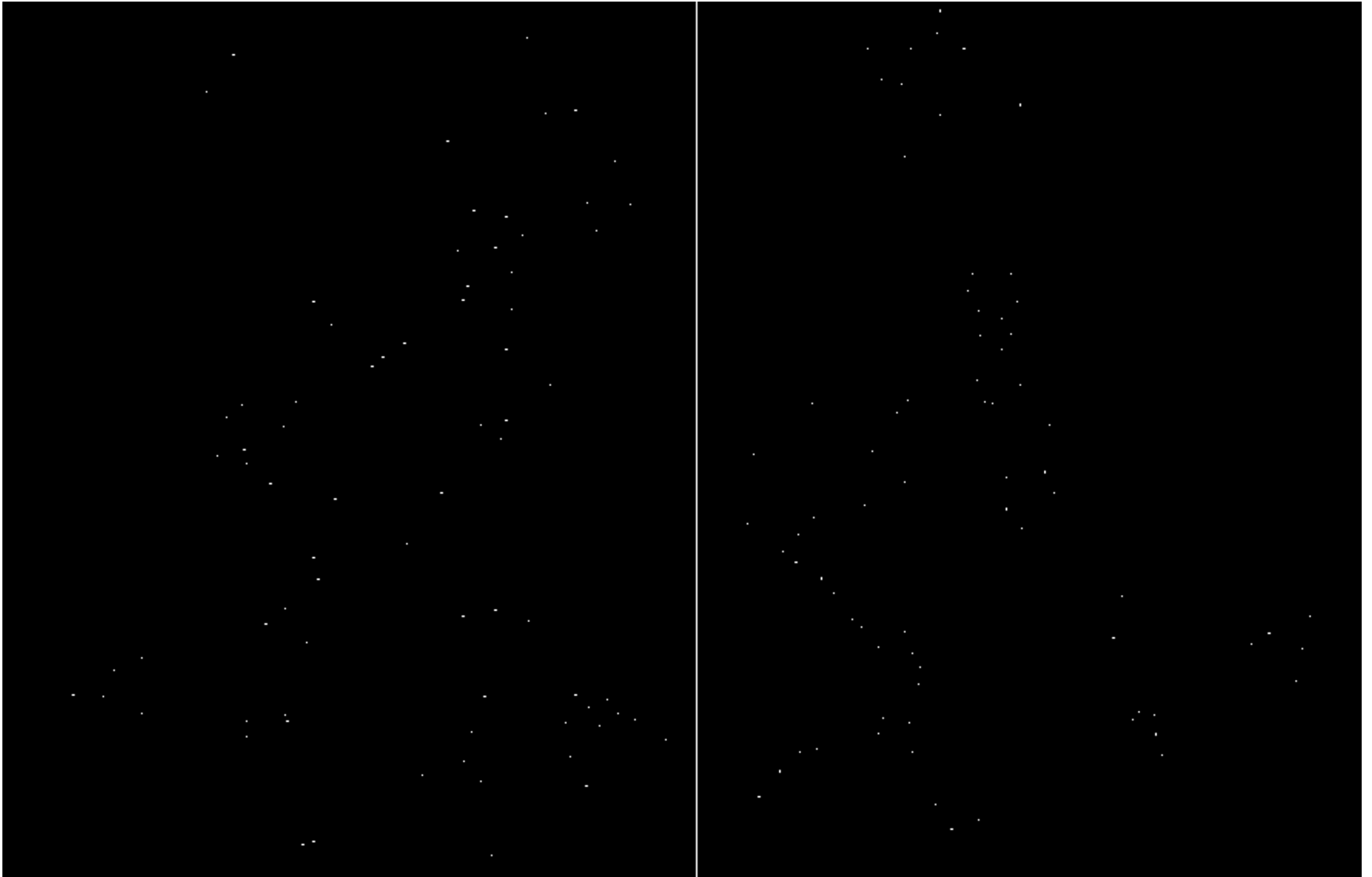
f value (red high, blue low)



Threshold ($f > \text{value}$)



Find local maxima of f



Harris features (in red)

